

NPM : 23312111

Nama : Muhammad Defrizal

Kelompok : Trio ORM

Dokumentasi Backend

Digi-Mount Rajabasa

1. Deskripsi Sistem

Digi-Mount Rajabasa merupakan aplikasi registrasi pendakian Gunung Rajabasa berbasis digital yang dibangun menggunakan arsitektur microservice secara bertahap.

Pada tahap pengembangan saat ini, sistem backend terdiri dari:

- API Gateway
- Jalur Service

Teknologi yang digunakan:

- Backend Framework: NestJS
 - Database: PostgreSQL
 - ORM: Prisma
 - API Style: REST API
 - Testing API: Apidog
-

2. Arsitektur Sistem

Struktur Backend

Client



API Gateway (Port 3003)



Jalur Service (Port 3001)



PostgreSQL Database

Penjelasan

API Gateway

API Gateway berfungsi sebagai:

- Entry point utama aplikasi
- Penghubung antara client dan service
- Penerus request ke service terkait
- Penanganan error terpusat

Jalur Service

Jalur Service bertanggung jawab terhadap:

- CRUD data jalur pendakian
 - Validasi data jalur
 - Akses database PostgreSQL menggunakan Prisma ORM
-

3. Struktur Project

```
api/  
├── gateway/  
└── jalur/
```

Gateway

```
gateway/src/  
├── common/  
│   └── instances/  
│       └── jalur.instance.ts  
└── jalur/  
    ├── dto/  
    ├── entities/  
    ├── jalur.controller.ts  
    ├── jalur.service.ts  
    └── jalur.module.ts
```

Jalur Service

```
jalur/src/  
├── generated/  
└── jalur/  
    ├── dto/  
    ├── jalur.controller.ts  
    └── jalur.service.ts
```

```
├── jalur.module.ts
├── prisma.service.ts
└── main.ts
```

4. Database Schema

Model Jalur

```
model Jalur {
  id          String   @id @default(uuid())
  nama        String   @unique
  deskripsi   String?
  tingkatKesulitan String
  createdAt   DateTime @default(now())
}
```

Penjelasan Field

Field	Tipe Data	Keterangan
id	String	Primary key menggunakan UUID
nama	String	Nama jalur pendakian
deskripsi	String?	Deskripsi jalur
tingkatKesulitan	String	Tingkat kesulitan jalur
createdAt	DateTime	Waktu data dibuat

5. Konfigurasi Port

Service	Port
Jalur Service	3001
API Gateway	3003

6. Endpoint API

Base URL Gateway

`http://localhost:3003/api`

Base URL Jalur Service

`http://localhost:3001/api`

7. Endpoint Jalur Service

7.1 Create Jalur

Request

POST /api/jalur

Body Request & Response Success

The screenshot shows the Apidog interface with a POST request to `http://localhost:3001/api/jalur`. The request body is in `x-www-form-urlencoded` format with the following parameters:

Name	Value	Type	Description
nama	Jalur Sumur Kumbang	string	
deskripsi	fasilitas basecamp lebih lengkap	string	
tingkatKesulitan	Menengah	string	

The response is in JSON format:

```
1 {
2   "success": true,
3   "message": "Jalur berhasil dibuat",
4   "data": {
5     "id": "8d6e8eb4-70be-4add-8959-660abe8b6f0f",
6     "nama": "Jalur Sumur Kumbang",
7     "deskripsi": "fasilitas basecamp lebih lengkap",
8     "tingkatKesulitan": "Menengah",
9     "createdAt": "2026-05-09T16:14:24.677Z"
10  }
```

The response status is 201, and the response body is 191ms and 489B. The response validation shows a success message: "HTTP Status Code should be 200".

Response Error

The screenshot shows the Apidog interface with a POST request to `http://localhost:3001/api/jalur`. The request body is in `x-www-form-urlencoded` format with the following parameters:

Name	Value	Type	Description
nama	Jalur Sumur Kumbang	string	
deskripsi	fasilitas basecamp lebih lengkap	string	
tingkatKesulitan	Menengah	string	

The response is in JSON format:

```
1 {
2   "message": "Jalur dengan nama yang sama sudah ada!",
3   "error": "Conflict",
4   "statusCode": 409
5 }
```

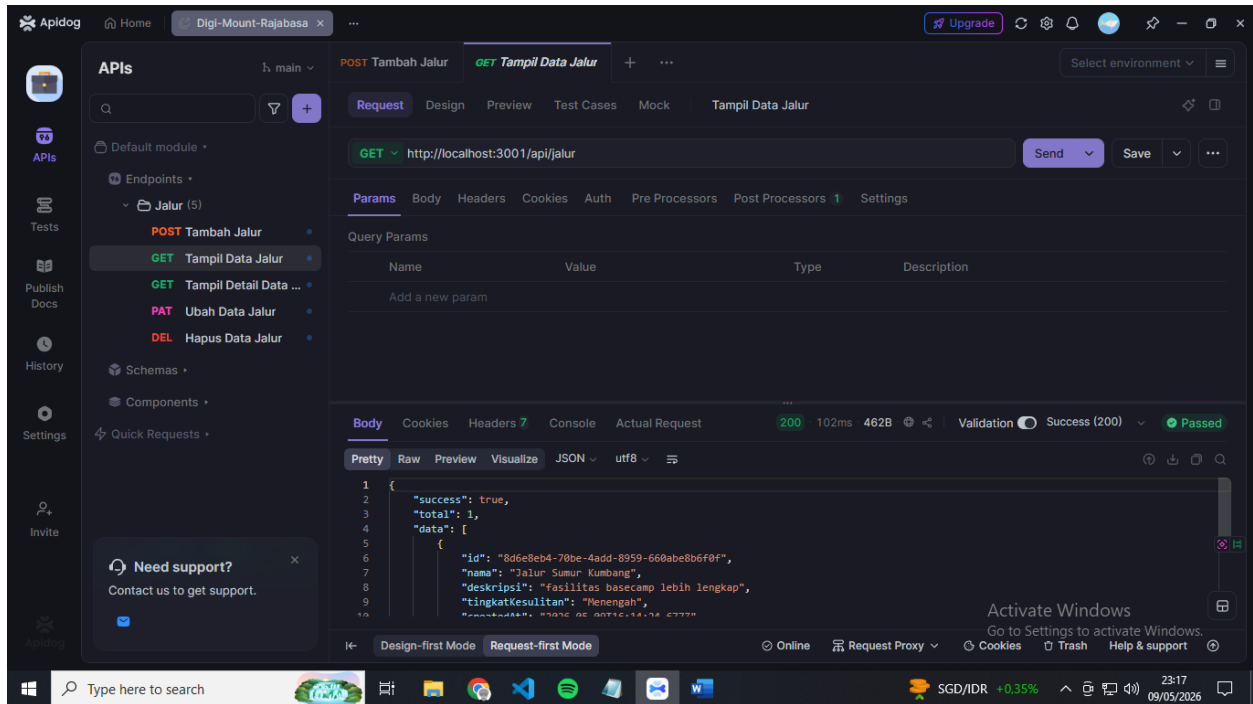
The response status is 409, and the response body is 51ms and 329B. The response validation shows an error message: "HTTP Status Code should be 200".

7.2 Get All Jalur

Request

GET /api/jalur

Response Success



7.3 Get Jalur By ID

Request

GET /api/jalur/:id

Response Success

The screenshot shows the Apidog interface with a successful GET request. The URL is `http://localhost:3001/api/jalur/8d6e8eb4-70be-4add-8959-660abe8b6f0f`. The response status is 200, with a response time of 247ms and a body size of 450B. The response is validated as 'Success (200)' and 'Passed'.

Body

```
1 {
2   "success": true,
3   "data": {
4     "id": "8d6e8eb4-70be-4add-8959-660abe8b6f0f",
5     "nama": "Jalur Sumur Kumb",
6     "deskripsi": "fasilitas b",
7     "tingkatKesulitan": "Menengah",
8     "createdAt": "2026-05-09T16:14:24.677Z"
9   }
10 }
```

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Activate Windows
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Response Error

The screenshot shows the Apidog interface with a 404 error response. The URL is `http://localhost:3001/api/jalur/udin`. The response status is 404, with a response time of 160ms and a body size of 320B. The response is validated as 'Success (200)' and 'Failed'.

Body

```
1 {
2   "message": "Data Jalur tidak ditemukan!",
3   "error": "Not Found",
4   "statusCode": 404
5 }
```

Response Validation

- HTTP Status Code should be 200

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Activate Windows
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7.4 Update Jalur

Request

PATCH /api/jalur/:id

Body Request & Response Success

The screenshot displays the Apidog API client interface. The left sidebar shows a list of endpoints under the 'Jalur' collection, including 'POST Tambah Jalur', 'GET Tampil Data Jalur', 'GET Tampil Detail Data ...', 'PATCH Ubah Data Jalur' (selected), and 'DEL Hapus Data Jalur'. The main panel shows the 'PATCH' request configuration for the endpoint 'http://localhost:3001/api/jalur/8d6e8eb4-70be-4add-8959-660abe8b6f0f'. The request body is set to 'x-www-form-urlencoded' and contains the following data:

Name	Value	Type	Description
nama	Jalur Teropong Kota Kalianda	string	
deskripsi	fasilitas basecamp lengkap loh guys	string	
tingkatKesulitan	Menengah	string	

The response is shown in the 'Body' tab, indicating a successful status (200) and a 'Passed' validation. The response body is a JSON object:

```
1 {
2   "success": true,
3   "message": "Jalur berhasil diperbarui",
4   "data": {
5     "id": "8d6e8eb4-70be-4add-8959-660abe8b6f0f",
6     "nama": "Jalur Teropong Kota Kalianda",
7     "deskripsi": "fasilitas basecamp lengkap loh guys",
8     "tingkatKesulitan": "Menengah",
9     "createdAt": "2026-05-09T16:14:24.677Z"
10  }
```

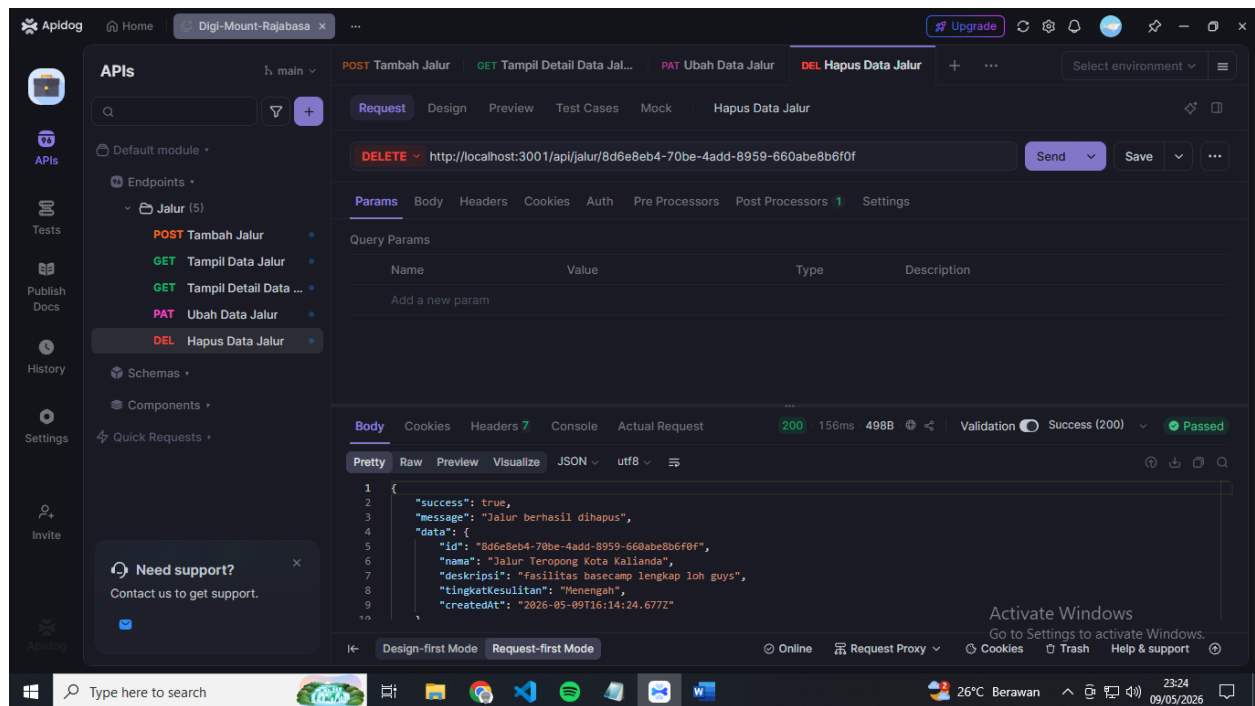
The bottom status bar shows the system time as 23:22 on 09/05/2026.

7.5 Delete Jalur

Request

DELETE /api/jalur/:id

Response Success



8. Endpoint Jalur Gateway

8.1 Create Jalur Gateway

Request

POST /api/jalur

Body Request & Response Success

The screenshot shows the Apidog interface with a POST request to `http://localhost:3003/api/jalur`. The request body is in `x-www-form-urlencoded` format with the following parameters:

Name	Value	Type	Description
nama	Jalur Desa Kecapi	string	
deskripsi	Jalur alternatif yang juga dikenal wa	string	
tingkatKesulitan	Menengah	string	

The response is a JSON object:

```
1 {
2   "success": true,
3   "message": "Jalur berhasil dibuat",
4   "data": {
5     "id": "4b24d666-24d7-4c7a-89c0-25ec70baba87",
6     "nama": "Jalur Desa Kecapi",
7     "deskripsi": "Jalur alternatif yang juga dikenal warga lokal",
8     "tingkatKesulitan": "Menengah",
9     "createdAt": "2026-05-09T16:32:46.273Z"
10  }
```

The response status is 201, and the response body is 414ms and 502B. The response validation shows a success message: "HTTP Status Code should be 200".

Response Error

The screenshot shows the Apidog interface with a POST request to `http://localhost:3003/api/jalur`. The request body is in `x-www-form-urlencoded` format with the following parameters:

Name	Value	Type	Description
nama	Jalur Desa Kecapi	string	
deskripsi	Jalur alternatif yang juga dikenal wa	string	
tingkatKesulitan	Menengah	string	

The response is a JSON object:

```
1 {
2   "message": "Jalur dengan nama yang sama sudah ada!",
3   "error": "Conflict",
4   "statusCode": 409
5 }
```

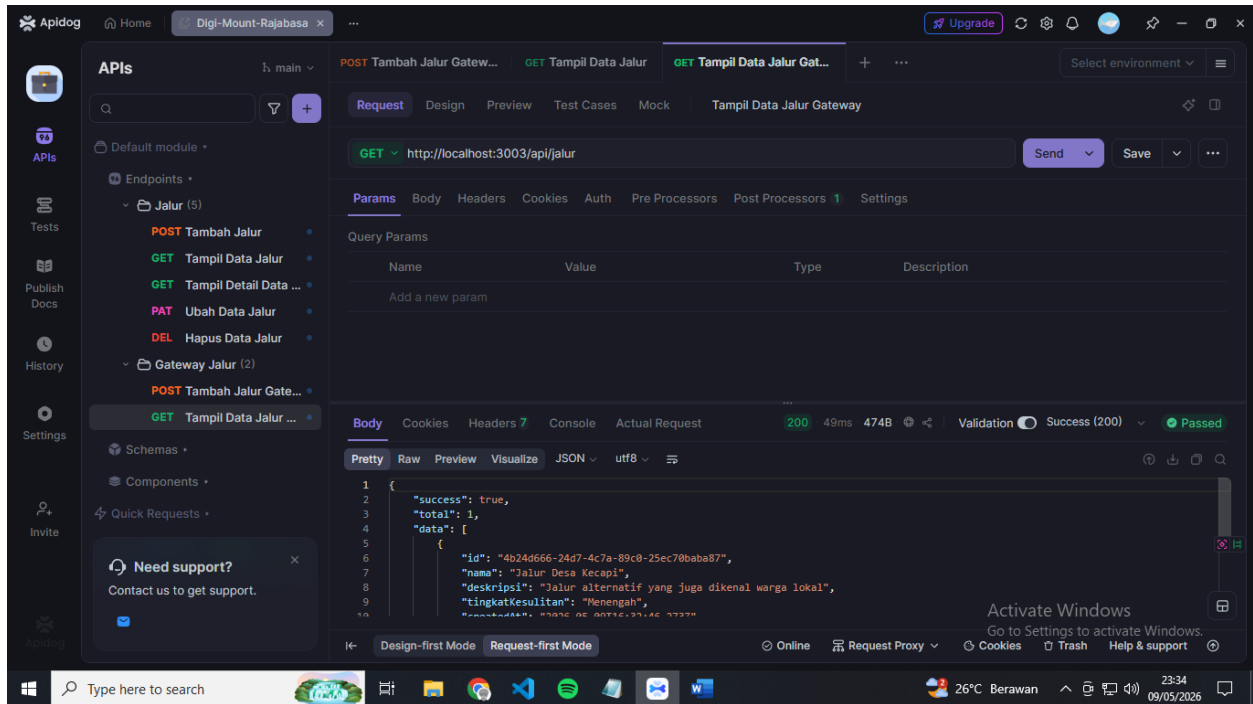
The response status is 409, and the response body is 223ms and 329B. The response validation shows an error message: "HTTP Status Code should be 200".

8.2 Get All Jalur Gateway

Request

GET /api/jalur

Response Success



8.3 Get Jalur By ID Gateway

Request

GET /api/jalur/:id

Response Success

The screenshot shows the Apidog interface with a successful GET request. The URL is `http://localhost:3003/api/jalur/4b24d666-24d7-4c7a-89c0-25ec70baba87`. The response status is 200, with a response time of 124ms and a body size of 462B. The response body is a JSON object:

```
1 {
2   "success": true,
3   "data": {
4     "id": "4b24d666-24d7-4c7a-89c0-25ec70baba87",
5     "name": "Jalur Desa Kecapi",
6     "deskripsi": "Jalur alternatif yang juga dikenal warga lokal",
7     "tingkatKesulitan": "Menengah",
8     "createdAt": "2026-05-09T16:32:46.273Z"
9   }
10 }
```

The interface also shows a sidebar with API endpoints and a bottom status bar with system information.

Response Error

The screenshot shows the Apidog interface with a 404 error response. The URL is `http://localhost:3003/api/jalur/iwan`. The response status is 404, with a response time of 56ms and a body size of 320B. The response body is a JSON object:

```
1 {
2   "message": "Data Jalur tidak ditemukan!",
3   "error": "Not Found",
4   "statusCode": 404
5 }
```

The interface also shows a sidebar with API endpoints and a bottom status bar with system information. A "Response Validation" panel on the right indicates that the HTTP Status Code should be 200.

8.4 Update Jalur Gateway

Request

PATCH /api/jalur/:id

Body Request & Response Success

The screenshot displays the Apidog API client interface. The left sidebar shows a list of APIs, with 'Jalur' selected. The main panel shows a PATCH request to the endpoint `http://localhost:3003/api/jalur/4b24d666-24d7-4c7a-89c0-25ec70baba87`. The request body is set to 'x-www-form-urlencoded' and contains the following data:

Name	Value	Type	Description
nama	Jalur Desa Kecapi	string	
deskripsi	biasanya lebih sepi guys	string	
tingkatKesulitan	Menengah	string	

The response is shown in the 'Body' tab, indicating a successful update:

```
{
  "success": true,
  "message": "Jalur berhasil diperbarui",
  "data": {
    "id": "4b24d666-24d7-4c7a-89c0-25ec70baba87",
    "nama": "Jalur Desa Kecapi",
    "deskripsi": "biasanya lebih sepi guys",
    "tingkatKesulitan": "Menengah",
    "createdAt": "2026-05-09T16:32:46.273Z"
  }
}
```

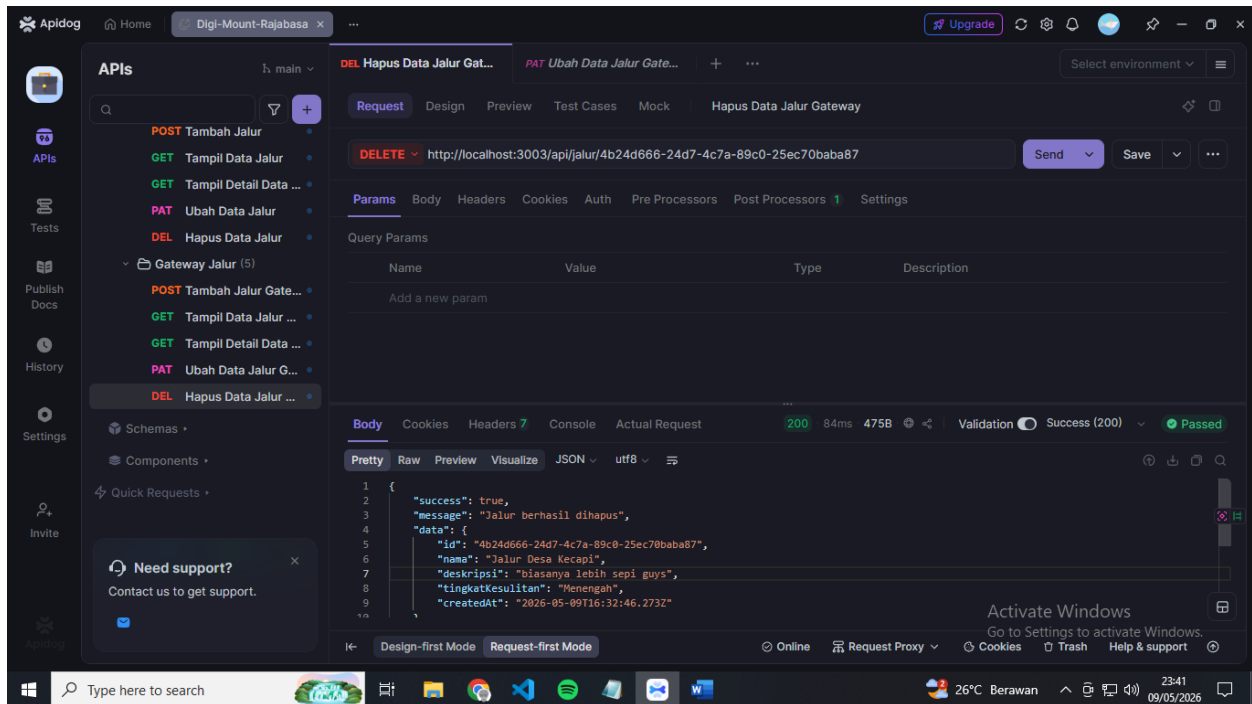
The status bar at the bottom shows 'Validation Success (200)' and 'Passed'.

8.5 Delete Jalur Gateway

Request

DELETE /api/jalur/:id

Response Success



9. Testing Menggunakan Apidog

Tujuan Testing

Pengujian API dilakukan menggunakan Apidog untuk memastikan:

- Endpoint berjalan dengan baik
- Validasi berjalan sesuai aturan
- Response API sesuai kebutuhan
- Error handling berjalan dengan benar

Skenario Testing

Endpoint	Method	Status
/api/jalur	POST	Berhasil
/api/jalur	GET	Berhasil
/api/jalur/:id	GET	Berhasil

Endpoint	Method	Status
/api/jalur/:id	PATCH	Berhasil
/api/jalur/:id	DELETE	Berhasil

10. Validasi dan Error Handling

Validasi yang Digunakan

- Nama jalur tidak boleh duplikat
- ID jalur harus valid
- Data jalur harus tersedia sebelum update/delete

Error Handling

Backend menggunakan exception bawaan NestJS:

Exception	Keterangan
ConflictException	Data duplikat
NotFoundException	Data tidak ditemukan
HttpException	Error dari gateway

11. Kesimpulan

Backend Digi-Mount Rajabasa telah berhasil mengimplementasikan:

- CRUD Jalur Pendakian
- API Gateway
- Komunikasi antar service menggunakan Axios
- Prisma ORM dengan PostgreSQL
- Error handling terpusat
- Struktur backend berbasis microservice secara bertahap

Tahap pengembangan selanjutnya:

- JWT Authentication
- Auth Service
- Pendaftaran Service
- Payment Service
- Logistik Service

- Role Authorization
- Dokumentasi Swagger